ORF 524 – Midterm Exam

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Contents

Notes:

- Duration of examination: *4 hours.*
  An additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
  Basic auxiliary information about standard distributions is provided at the end of the exam.
  This is otherwise a closed-book and individual examination.
  Do not consult textbooks, online materials, or other sources.
  Do not consult with classmates, other people or in online forums.

1 Question 1: Probability Theory (25 points)

Let $(X_1, X_2, \ldots, X_n)$ be independent (but not necessarily identically distributed) random variables with $\mu = \mathbb{E}[X_i]$ and $\sigma_i^2 = \mathbb{V}[X_i]$ for $i = 1, 2, \ldots, n$ (all expectations are assumed to exist). A random variable Y is said to be *sub-Gaussian* if there is a positive parameter γ such that

$$\mathbb{E}\big[\exp\big(t(Y - \mathbb{E}[Y])\big)\big] \leq \exp\Big(\frac{\gamma^2 t^2}{2}\Big), \qquad \text{for all } t \in \mathbb{R}.$$

We explore this property and use it to construct valid finite-sample and distribution-free confidence intervals for the common mean μ based on the sample average $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$.

1. Let $X_i \sim \mathsf{Normal}(\mu, \sigma_i^2)$ for $i = 1, 2, \ldots, n$. Show that $\bar{X}$ is sub-Gaussian, and identify its parameter γ.

2. Let X_i be sub-Gaussian with parameter γ_i for $i = 1, 2, \ldots, n$.

   (a) Show that $\bar{X}$ is sub-Gaussian, and identify its parameter.

   (b) Show that

   $$\mathbb{P}[\bar{X} - \mu \geq x] \leq \inf_{t>0} \exp\Big(\frac{\widetilde{\gamma}^2 t^2}{2n} - tx\Big) = \exp\Big(-\frac{nx^2}{2\widetilde{\gamma}^2}\Big), \qquad \text{for all } x \in \mathbb{R},$$

   where $\widetilde{\gamma} = \sqrt{\frac{1}{n}\sum_{i=1}^n \gamma_i^2}$. Use this result to prove the (two-sided) Hoeffding Inequality:

   $$\mathbb{P}[|\bar{X} - \mu| \geq x] \leq 2\exp\Big(-\frac{nx^2}{2\widetilde{\gamma}^2}\Big).$$

3. Show that if a random variable W satisfies

   $$\mathbb{P}[|W| \geq w] \leq a\exp\left(-\frac{bw^2}{c + dw}\right), \qquad \text{for all } w > 0,$$

   where $a, b > 0$ and $c, d \geq 0$, then

   $$\mathbb{P}\left[|W| \leq \sqrt{\frac{c}{b}\ln\Big(\frac{a}{\alpha}\Big)} + \frac{d}{b}\ln\Big(\frac{a}{\alpha}\Big)\right] \geq 1 - \alpha,$$

   for $\alpha \in (0, 1)$.

4. Assuming that X_i is sub-Gaussian with *known* parameter γ_i for $i = 1, 2, \ldots, n$, propose a confidence interval for μ with a coverage probability of at least 95%.

2 Question 2: Statistical Decision Theory (25 points)

Let $\mathbf{X} \sim \mathbb{P}_\theta$ be a random vector with $\mathbb{P}_\theta \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$. Assume that $T = T(\mathbf{X})$ is a complete sufficient statistic for $\mathcal{P}$ and that $V = V(\mathbf{X})$ is an ancillary statistic for $\mathcal{P}$.

1. Show that $p_A = \mathbb{E}_\theta[\mathbb{1}(V \in A)]$ and $q_A(T) = \mathbb{E}_\theta[\mathbb{1}(V \in A) \mid T]$ are not functions of θ, for all Borel measurable sets A.

2. Show that $p_A = \mathbb{E}_\theta[q_A(T)]$, and that $p_A = q_A(T)$ almost surely.

3. Show that $\mathbb{P}_\theta(T \in B, V \in A) = \mathbb{P}_\theta(T \in B)\mathbb{P}_\theta(V \in A)$ for all Borel measurable sets A and B.

4. Let $\mathbf{X} = (X_1, \dots, X_n)$ be an i.i.d. random sample from $\mathsf{Normal}(\mu, \sigma^2)$, with $\sigma > 0$ known. Let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \text{and} \qquad \bar{X}(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} w_i X_i,$$

   for some non-random weights $\mathbf{w} = (w_1, \dots, w_n)'$.

   (a) Show that $\bar{X}(\mathbf{w})$ is an ancillary statistic for μ if and only if $\sum_{i=1}^{n} w_i = 0$.

   (b) Provide sufficient conditions for $\bar{X}$ and $\bar{X}(\mathbf{w})$ to be independent.

3 Question 3: Finite-Sample Statistical Inference (50 points)

Let $(X_i : 1 \le i \le n)$ be an i.i.d. random sample from $X \sim \mathbb{P}_\theta$, with absolutely continuous d.f.

$$F(x;\theta) = \mathbb{P}_\theta(X \le x) = (a + bx^{-1/\theta})\mathbb{1}\{x > 2\},$$

where $\theta \in \Theta = (0,1)$ is an unknown parameter, and a and b do not depend on x.

1. Consider basic distributional properties of $\mathbb{P}_\theta$.
   (a) Find the values of a and b.
   (b) Find the Lebesgue density of $\mathbb{P}_\theta$, denoted $f(\cdot;\theta)$. Is $\{\mathbb{P}_\theta : \theta \in \Theta\}$ an exponential family?

2. Consider point estimation of θ using the following two estimators:

$$\hat{\theta}_{\mathtt{MM}} = \frac{\bar{X} - 2}{\bar{X}} \qquad \text{and} \qquad \hat{\theta}_{\mathtt{ML}} = \frac{1}{n}\sum_{i=1}^{n} \log\left(\frac{X_i}{2}\right),$$

   where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. The estimators $\hat{\theta}_{\mathtt{MM}}$ and $\hat{\theta}_{\mathtt{ML}}$ correspond to the method of moments and maximum likelihood estimators respectively.

   (a) Compute $\mathbb{E}_\theta[X_i]$. Characterize the direction of bias (if any) of $\hat{\theta}_{\mathtt{MM}}$: is $\hat{\theta}_{\mathtt{MM}}$ on average an overestimate or underestimate of θ?
   (b) Show that $\log(X_i/2) \sim \mathsf{Exponential}(\theta)$.
   Compute the bias of $\hat{\theta}_{\mathtt{ML}}$ for θ. If biased, characterize the direction.
   (c) Determine whether $\hat{\theta}_{\mathtt{MM}}$ is based on a complete sufficient statistic.
   Determine whether $\hat{\theta}_{\mathtt{ML}}$ is based on a complete sufficient statistic.
   (d) Compute the Cramér-Rao lower bound for any (regular) unbiased estimator of θ.
   Determine whether $\hat{\theta}_{\mathtt{MM}}$ is a UMVU estimator.
   Determine whether $\hat{\theta}_{\mathtt{ML}}$ is a UMVU estimator.
   In each case, verify whether the Cramér-Rao lower bound is attained.

3. Consider the following one-sided hypothesis testing problem:

$$\mathsf{H}_0 : \theta \ge \theta_0 \quad \text{vs.} \quad \mathsf{H}_1 : \theta < \theta_0, \qquad \text{for some fixed } \theta_0 \in (0,1).$$

   (a) Find the uniformly most powerful (UMP) testing procedure of level $\alpha \in (0,1)$.
   (b) Construct an exact $(1-\alpha)$ confidence interval for θ.
   (c) Find the power function $\beta_n(\theta)$ of the UMP test and sketch its graph.
   (d) For each $\theta \in (0,1)$, find $\lim_{n\to\infty} \beta_n(\theta)$.

   Hint: You may use the following two facts without proof.

   (i) If $W_1, \ldots, W_n$ are i.i.d. $\mathsf{Exponential}(\lambda)$, then $\sum_{i=1}^{n} W_i \sim \mathsf{Gamma}(n,\lambda)$.
   (ii) If $Y_n \sim \mathsf{Gamma}(n,\lambda_n)$, then $\lim_{n\to\infty} \mathbb{P}\left(\frac{Y_n - n\lambda_n}{\sqrt{n}\lambda_n} \le x\right) = \Phi(x)$ for all $x \in \mathbb{R}$, where Φ is the c.d.f. of the standard normal distribution.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m$, $x = a_1, ..., a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, \ \ t \in \mathcal{R}$
$DU(a_1, ..., a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m$, $\bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}$, $m = 1, 2, ...$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, \ \ x = 0, 1, ..., n$
	m.g.f.	$(pe^t + 1 - p)^n, \ \ t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], \ \ n = 1, 2, ...$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, \ \ x = 0, 1, 2, ...$
	m.g.f.	$e^{\theta(e^t - 1)}, \ \ t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1} p, \ \ x = 1, 2, ...$
	m.g.f.	$pe^t/[1 - (1-p)e^t], \ \ t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
geometric		$x = 0, 1, ..., \min\{r, n\}, \ \ r - x \leq m$
	m.g.f.	No explicit form
$HG(r, n, m)$	Expectation	rn/N
	Variance	$rnm(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, ..., \ N = n + m$
Negative	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, \ \ x = r, r+1, ...$
binomial	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, \ \ t < -\log(1-p)$
	Expectation	r/p
$NB(p, r)$	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], \ \ r = 1, 2, ...$
Log-	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, \ \ x = 1, 2, ...$
distribution	m.g.f.	$\log[1 - (1-p)e^t]/\log p, \ \ t \in \mathcal{R}$
	Expectation	$-(1-p)/(p \log p)$
$L(p)$	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt}-e^{at})/[(b-a)t],\ \ t\in\mathcal{R}$
$U(a,b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a,\ b\in\mathcal{R},\ \ a<b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t+\sigma^2t^2/2},\ \ t\in\mathcal{R}$
$N(\mu,\sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1},\ \ t<\theta^{-1}$
$E(a,\theta)$	Expectation	$\theta+a$
	Variance	θ^2
	Parameter	$\theta>0,\ \ a\in\mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2},\ \ t<1/2$
χ^2_k	Expectation	k
	Variance	$2k$
	Parameter	$k=1,2,...$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha},\ \ t<\gamma^{-1}$
$\Gamma(\alpha,\gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma>0,\ \alpha>0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha,\beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha>0,\ \ \beta>0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t-\sigma\lvert t\rvert}$
$C(\mu,\sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, $(n>1)$
	Variance	$n/(n-2)$, $(n>2)$
	Parameter	$n=1,2,...$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, $(m>2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m>4)$
	Parameter	$n=1,2,...$, $m=1,2,...$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x-\mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu,\sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
	Parameter	$\mu\in\mathcal{R}$, $\sigma>0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha,\theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1}+1)$
	Variance	$\theta^{2/\alpha}\left\{\Gamma(2\alpha^{-1}+1)-\left[\Gamma(\alpha^{-1}+1)\right]^2\right\}$
	Parameter	$\theta>0$, $\alpha>0$
Double	p.d.f.	$\frac{1}{2\theta}e^{-\lvert x-\mu\rvert/\theta}$
Exponential	m.g.f.	$e^{\mu t}/(1-\theta^2t^2)$, $\lvert t\rvert<\theta^{-1}$
	Expectation	μ
$DE(\mu,\theta)$	Variance	$2\theta^2$
	Parameter	$\mu\in\mathcal{R}$, $\theta>0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a,\theta)$	Expectation	$\theta a/(\theta-1)$, $(\theta>1)$
	Variance	$\theta a^2/[(\theta-1)^2(\theta-2)]$, $(\theta>2)$
	Parameter	$\theta>0$, $a>0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1+e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1+\sigma t)\Gamma(1-\sigma t)$, $\lvert t\rvert<\sigma^{-1}$
$LG(\mu,\sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu\in\mathcal{R}$, $\sigma>0$